Exercises 7

Exercise 1 (Fixed points of real functions)**.** Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = x^2 - x - 3 \ .$$

Describe the set $F_f \subseteq \mathbb{R}$, the set of fixed points of f. Does f admit a least, or respectively greatest, fixed point?

Exercise 2 (Iterating sp and wp)**.** This question is inspired by relationships between state sets in a transition system.

Let S be a non-empty set of states and $r \subseteq S \times S$ a relation. As usual, define

- for every $P \subseteq S$, $sp(P, r) = \{x' \mid \exists x \in P.\ (x, x') \in r\}$
- for every $Q \subseteq S$, $wp(r, Q) = \{x \mid \forall x'.\ (x, x') \in r \to x' \in Q\}$

Let $2^S = \{X \mid X \subseteq S\}$ be the set of all sets of states. Let $\mathit{Init} \subseteq S$ and $\mathit{Good} \subseteq S$. Define:

- $F : 2^S \to 2^S$ as $F(X) = \mathit{Init} \cup sp(X, r)$
- $B : 2^S \to 2^S$ as $B(Y) = \mathit{Good} \cap wp(r, Y)$
- $l = \bigcup_{i \geq 0} F^i(\emptyset)$
- $h = \bigcap_{i \geq 0} B^i(S)$

Consider the partial order $(2^S, \subseteq)$. For each of the following properties on this order, determine if it holds or not. Provide a proof or a counterexample in each case.

1. F is monotonic
2. B is monotonic
3. l is the least fixed point of F.
4. h is the greatest fixed point of B.
5. $l \subseteq h$.
6. $l \subseteq \mathit{Good}$ implies $\mathit{Init} \subseteq h$
7. $\mathit{Init} \subseteq h$ implies $l \subseteq \mathit{Good}$

Exercise 3 (Fixed points in lattices)**.** Consider a complete lattice with the set of elements L and the ordering relation $\sqsubseteq: L \times L$. Recall that in a complete lattice, every set $X \subseteq L$ has a least upper bound and a greatest lower bound in L.

Let $G : L \to L$ be a monotonic function with respect to $\sqsubseteq$. Let

$$Fix = \{x \in L \mid G(x) = x\}$$

be the set of all fixed points. Let $x, y \in Fix$ be two fixed points. Prove or disprove:

1. $x \sqcup y \sqsubseteq G(x \sqcup y)$
2. $G(x \sqcup y) \sqsubseteq x \sqcup y$
3. $x \sqcup y \in Fix$
4. Let $B = \{b \in Fix \mid x \sqsubseteq b \wedge y \sqsubseteq b\}$. Then B has a least element, that is, there exists an element $z \in B$ such that $\forall b \in B.\ z \sqsubseteq b$. (Possibly difficult)

Exercise 4 (Abstract interpretation)**.** Consider a program on two integer variables, that is the set of sets of states, i.e. the concrete domain, is $C = 2^{\mathbb{Z} \times \mathbb{Z}}$. We want to represent these states with an abstract domain A such that a state is represented by exact options for the first variable, and only as an interval for the second variable.

1. Express the lattice A corresponding to this domain.
2. Give expressions for monotonic functions $\alpha : C \to A$ and $\gamma : A \to C$ such that they form a (non-trivial) Galois connection.

Exercise 5 (Program analysis)**.** Consider a program that manipulates two integer variables x and y. Consider any assignment of the form $x := e$, where e is a (normalized) linear combination of integer variables, for example

$$x := 2 * x - 5 * y\ .$$

Consider an abstract analysis with independent intervals for both variables. Describe an algorithm that, given the expression e, and intervals for x (noted $[a_x, b_x]$) and y (noted $[a_y, b_y]$) computes the new interval $[a, b]$ for x after the assignment statement.

Provide an algorithm that takes as input e, and the intervals for x and y as below, and produces the new interval $[a, b]$ for x after the assignment statement. You may disregard integer overflows and underflows.

```scala
import math.*

// represents a*x + b*y + c
case class LinearExpr(a: Int, b: Int, c: Int)
case class Interval(min: Int, max: Int)

// x := e under given intervals for x and y
def nextInterval(
    e: LinearExpr,
    x: Interval,
    y: Interval
): Interval =
  // your code here
  ???
```